PRINCE

Spatial Data Science & Earth Observation Researcher

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Profile

Undergraduate researcher in Geography at the University of Delhi, specializing in Remote Sensing, GIS, and computational terrain analysis. My academic research focuses on Himalayan disaster vulnerability, including glacial lake outburst floods (GLOF) and rural community accessibility. Experienced in processing optical and radar satellite data, developing open-source tools with Python and QGIS, and conducting field-based spatial surveys. Dedicated to open science and aiming to pursue graduate research in Earth Observation, Geoinformatics, and Cartography

Education

University of Delhi – Dr. Bhim Rao Ambedkar College *Expected July 2027*

BA (Hons.) Geography with Research (4-Year Track) | First Class Standing *New Delhi, India*

- **Core Quantitative Modules:** Remote Sensing & GIS, Geostatistics, Cartography & Geovisualization, Environmental Geomorphology, Terrain Modeling, Quantitative Research Methodologies.

Research & Publications

AI-Based Monitoring and Future Prediction of High-Altitude Glacial Lakes for GLOF Risk Assessment in Himachal Pradesh *Ongoing Thesis*

Undergraduate Dissertation | Western Himalayas *Himachal Pradesh, India*

- Architecting an Earth Observation monitoring pipeline combining **Sentinel-1 SAR** dual-polarization backscatter (VV/VH) and **Sentinel-2 multispectral imagery** (NDWI/MNDWI indices) to penetrate chronic cloud cover and track moraine-dammed glacial lake expansion.
- Implementing deep learning segmentation to detect lake boundary fluctuations and predict sudden outburst flood risk across high-altitude catchments.

Community Isolation Risk along the NH-7 Himalayan Transport Corridor *Sept 2026*

Independent Lead Spatial Researcher Uttarakhand, India **Official Citable Record:** Zenodo DOI: 10.5281/zenodo.23670659 | **Esri India User Conference 2026 Selection**

- Formulated an independent spatial gravity model evaluating village accessibility exposure across **3,419 rural settlements** along the NH-7 arterial corridor within a 15 km buffer.
- Integrated **ESA Copernicus GLO-30 DEM** (slope gradient and elevation), OpenStreetMap friction surfaces, and Census demographics using an Analytic Hierarchy Process (AHP) multi-criteria framework: **Road Accessibility (46.91%)**, **Healthcare Proximity (20.10%)**, **NH-7 Proximity (20.10%)**, **Slope Gradient (8.62%)**, and **Elevation (4.27%)**.
- Conducted in-situ hazard ground-truthing in Rudraprayag (Mastura landslide corridor); published full open-access citable dataset on CERN/Zenodo.

Systems Architecture & Professional Experience

CamAI – Real-Time Visual Intelligence Ecosystem *Jul 2026 – Present*

*Founder & Chief Systems Architect | Live Platform: **camai.princesite.in*** *Delhi, India*

- Designed and architected an end-to-end vision AI platform (camai.princesite.in) for intelligent edge surveillance, real-time object tracking, and spatial telemetry.
- Orchestrated the inference pipeline utilizing **TensorRT FP16**, **ByteTrack Re-ID**, and custom Python/FastAPI microservices connected to an Electron/TypeScript dashboard; currently in enterprise acquisition negotiations.

Independent Geospatial Consultant *Jan 2025 – Present*

Lead Consultant & Spatial Analyst *Remote / Client-Facing*

- **Post-Mining Solar Reclamation (Confidential Client Project):** Conducted multi-criteria suitability modeling for a decommissioned open-cast coal mine to evaluate land repurposing for utility-scale solar farms. Modeled the Cost Efficiency Index (CEI) and Solar Potential Index (SPI) to determine composite suitability and identify optimal candidate parcels for clean energy transition.
- **Utility Solar Siting (Pune):** Executed Multi-Criteria Decision Making (MCDM) across **20+ geospatial layers** (solar irradiance, grid proximity, slope, land-use zoning) for utility-scale solar siting.
- **Healthcare Accessibility (Ludhiana):** Modeled travel-time isochrones and service-area catchments to identify optimal emergency medical facility placement.

Carbon X Services Pvt. Ltd. *Mar 2026 – Jun 2026*

GIS Analyst | Industrial Decarbonisation & CCUS *Remote*

- Conducted geological and spatial suitability modeling for underground Carbon Capture and Storage (CCS) to sequester industrial emissions from heavy steel manufacturing facilities.
- Integrated multi-criteria geospatial layers including deep geological basin integrity, fault line/seismic risk buffers,

proximity to steel emission clusters, and land-use restrictions to delineate optimal subsurface injection zones.

Terra AI – Autonomous Spatial Intelligence Engine *Dec 2025 – Apr 2026*
Founder & Lead Developer | Completed R&D Proof-of-Concept *Delhi, India*

- Architected a proof-of-concept autonomous spatial AI engine exploring **Multi-Agent LLM swarms** for geospatial ETL automation, coordinate re-projection, and raster algebra execution.

Open-Source Contributions & Software

AutoGIS (v1.0.3) – Autonomous GIS Assistant & GEE Tool Suite *Oct 2026*
*Creator & Lead Systems Architect | Live: **autogis.princesite.in** | QGIS Plugin ID: 6782* *Open Source*

- Architected an autonomous QGIS plugin featuring **39 autonomous spatial tools** integrated with **Google Earth Engine (GEE)** for automated raster algebra, cloud masking, and spatial ETL workflows.
- Submitted to the official QGIS Plugin Repository (Plugin ID: 6782); deployed dedicated release portal (autogis.princesite.in) under the MIT License.

GeoLibre – India GIS Open Dataset Ecosystem *Oct 2026*
*Core Open-Source Contributor | **8.3k+ GitHub Stars Ecosystem*** *Open Source*

- Authored and integrated an accepted core pull request (14 files, ∼1,400 lines) enabling regional administrative querying and automated GeoJSON streaming by resolving Git LFS pointer and CORS bottlenecks.
- Implementation officially selected and indexed as a benchmark reference solution in the **XERJ Public Corpus Hub** for autonomous coding agents and developers worldwide.

Technical & Architectural Skills

Earth Observation & GIS: Copernicus DEM GLO-30, Sentinel-1 SAR, Sentinel-2 Optical, Landsat 8/9, Google Earth Engine (GEE), ArcGIS Pro, QGIS, ESA SNAP, GDAL, GeoPandas.

Spatial Modeling: AHP Multi-Criteria Decision Making (MCDM), Solar Siting Indices (CEI, SPI), Composite Suitability Modeling, Underground CCS Siting, Terrain Geomorphometry.

Systems & AI Architecture: Multi-Agent LLM Systems (Terra AI), AI-Assisted Software Synthesis, Edge Inference (TensorRT, ByteTrack), FastAPI, React/TypeScript.

Research Methodologies: ICMR Exposure Modeling, In-Situ GPS Ground-Truthing, Stratified Sampling, Open Science Workflows (Zenodo, ORCID).

Research Training & Certifications

- **ICMR & ESDA India:** Capacity Building Workshop on Environmental Health, Exposure Assessment, and Data-Driven Modelling (Hands-on Scientific Training supported by Ministry of Health and Family Welfare, Govt. of India).
- **United Nations University (UNU-IAS):** Net Zero 101 – Carbon Policy, Measurement & Climate Modeling (2026).
- **The Open University (OpenLearn):** Climate Change and Renewable Energy Spatial Modeling (2026).
- **Esri India User Conference 2026:** National Call for Maps Selection & Showcase Certificate.

Field Research Expeditions & Leadership

- **Landslide Risk Ground-Truthing (Rudraprayag, Uttarakhand):** Executed in-situ hazard mapping and GPS sampling along the Mastura landslide corridor to evaluate community isolation and disaster vulnerability in the Himalayan region.
- **Arid Socio-Economic Survey (Jaisalmer, Rajasthan):** Administered GPS spatial sampling and field demographic surveys to analyze rural demographic adaptation in arid ecosystems.
- **Academic Technical Exhibits Leadership:** Headed technical coordination, digital spatial exhibits, and visual conference communications for University Geo Fests (BHŪ-CHETNA 2024, 2025).